

# GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

## Internal Examination

Level: Bachelor Semester: Fall Year: 2023  
Programme: BE software Full Marks: 100  
Course: Microprocessor and Computer Architecture Pass Marks: 45  
Time: 3 hrs

*Candidates are required to give answers in their own words as far as practicable. Figures in the margin indicate full marks.*

### Attempt all the questions:

1. a) Define the terms: Microprocessor and Microcontroller. Differentiate between Von Neumann Architecture and Harvard Architecture. 7  
b) Explain, in brief, the 8085 microprocessor with its block diagram. 8
2. a) Write an assembly language program for 8085 MP to add two 8-bit numbers (F1 and 15), stored at C050H and C051H, with carry. 7  
b) What is instruction cycle and machine cycle? Draw timing diagram for the instruction MOV A, B. 8
3. a) How 8086 MP is advance than 8085 MP? Explain the addressing modes in 8086 Microprocessor. 7  
b) Design a very simple CPU, with its state diagram, register section and ALU, for the given set of instructions. 8

| Instruction | Instruction Code | Operation                          |
|-------------|------------------|------------------------------------|
| JMP1        | 00AAAAAA         | $PC \leftarrow AAAAAA + 1$         |
| INC2        | 01XXXXXX         | $AC \leftarrow AC + 2$             |
| ADD1        | 10AAAAAA         | $AC \leftarrow AC + M[AAAAAA] + 1$ |
| SKIP        | 11XXXXXX         | $PC \leftarrow PC + 1$             |

4. a) Design hardwired control unit for the CPU described in question 3b. Also, briefly explain its operation. 7  
b) Design a micro-sequencer control unit with horizontal microcode for the CPU described in question 3b. 8

### OR

Differentiate between:

- i) Horizontal microcode and Vertical microcode
- ii) Hardwired CU and Microprogrammed CU

5. a) Define RTL. Write the RTL codes for Booth's algorithm. Trace with RTL to multiply -12 and 4. 7  
b) What is memory hierarchy? Discuss briefly on different cache mapping techniques. 8
6. a) What is the significance of DMA? Explain its working phenomenon with a block diagram. 7  
b) Explain Instruction pipeline with an example. In a system, there are 5 register windows. Each of the window has 4 input register and 4 output register and 10 local registers. The system has 20 global registers. Calculate the total no. of registers. Show the pictorial representation of this register windowing architecture. 8

### OR

What is Flynn's taxonomy? Explain UMA and NUMA architectures.

7. Write short notes on: (any two) 2x5
  - a) Micro-sequencer
  - b) Floating point number representation
  - c) UART
  - d) Multicore architecture

**POKHARA UNIVERSITY****Internal Assessment**

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Semester: Spring

Year: 2023

Program: B.E. Software

Full Marks: 100

Subject: Microprocessor and Computer Architecture

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**Attempt all the questions.**

- 1 a What do you understand by the term microprocessor? Highlight the brief history of generation of microprocessor. 7
- b What are the functional units of 8085 microprocessor? Explain with the block diagram of 8085 microprocessor. 8
- 2 a What are addressing modes? Explain the types of addressing modes available in 8085 microprocessor with examples. 7
- Or
- Why do you need instruction cycle in 8085 microprocessor? Define t-states, timing diagram, machine cycle briefly.
- b Draw a timing diagram for the given instruction MVI B, 43 8
- 3 a Write an assembly language program for the subtraction of two 8-bit numbers using 8085 microprocessor along with the flowchart. 8
- b Explain the role of flag register in 8086 microprocessor? 7
- Or
- Give the technical difference between the 8085 and 8086 microprocessor. How does both help in our daily life?
- 4 a There is the very simple CPU for the given set of instructions. 8

| Instruction | Instruction Code | Operation         |
|-------------|------------------|-------------------|
| JMP1        | 00AAAAAA         | PC←AAAAAA+1       |
| INC2        | 01XXXXXX         | AC←AC+2           |
| ADD1        | 10AAAAAA         | AC←AC+M[AAAAAA]+1 |
| SKIP        | 11XXXXXX         | PC←PC+1           |

Let the instruction width be 8 bits and address bit is 6 bits. Design a CPU state diagram, register section and the ALU.

- b Design a hardwired control unit for the same CPU designed in the above question 4a. 7
- 5 a Design the micro sequencer control unit with horizontal microcode for the CPU described in the question 4a. 8
- b Write the RTL code for the Booth's Algorithm. 7
- Use it to multiply  $-3 * -5$
- 6 a What is cache memory? Explain the working mechanism of direct mapping with the example. 8
- Or
- What are the different types of I/O techniques? Explain them.
- b What is the register window? In a system there are 5 windows of registers each registers share 4 input register and 4 output registers. Each of window has 10 local registers. The system has 20 global registers. Calculate the total number of registers in the system. Show the pictorial representation as well. 7
- 7 Write short notes on any two. 2\*5= 10
- a Ring vs Mesh topology
- b BCD
- c Lookup ROM
- d Multicore architecture